

Lecture Note
(Review) Simple Regression Analysis II

(A) Sampling Distributions of the OLS Estimators

(B) Confidence Intervals

(C) Empirical Examples (Estimation, Interpretation, Testing)

(A) Sampling Distributions of the OLS Estimators

- **ASSUMPTIONS MLR. 6 (Normality):** The population error u is independent of the explanatory variables $x_1, \dots, x_k$, and is normally distributed with zero mean and variance σ^2 : $u \sim \text{Normal}(0, \sigma^2)$.
- **Discussion:**
 - (a) This is a stronger assumption than Assumptions MLR. 4 and MLR.5.
Assumption MLR. 4 is about the first moment of u , $E(u|x_1, \dots, x_k) = E(u) = 0$ and Assumption MLR. 5 is about the second moment of u , $\text{Var}(u|x_1, \dots, x_k) = \text{Var}(u) = \sigma^2$. So, Assumption MLR. 6 **implies** Assumptions MLR 4 and 5.

(b) For cross-sectional regression applications, the Assumptions MLR.1 through MLR.6 are called the **classical linear model (CLM) assumptions**. Thus, we will refer to the model under these six assumptions as the **classical linear model**. It is best to think of the CLM assumptions as containing all of the Gauss-Markov assumptions *plus* the assumption of a **normally** distributed error term.

- **Theorem: Normal Sampling Distributions:** Under Assumptions 1 through 6, conditional on the sample values of the explanatory variables,

$$\hat{\beta}_j \sim N\left(\beta_j, \frac{\sigma^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}\right).$$

Equivalently,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}}} \sim N(0,1)$$

B. Confidence Intervals

- Given the sampling distribution of OLS estimators $\hat{\beta}_j$, we can show how **accurate** the estimate is (inference).

Example: A regression model of return to education

$$\begin{aligned}\log(wage) = & \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married \\ & + \beta_5 black + \beta_6 south + \beta_7 urban + u\end{aligned}$$

Suppose we are interested in the partial effect of education on wage: β_1 . For a particular sample (e.g., WAGE2), the OLS estimate $\hat{\beta}_1$ provides **one** point estimate for β_1 .

Example: An empirical Result based on WAGE2.dta data set.

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. reg lwage educ exper tenure married black south urban
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Source	SS	df	MS	Number of obs = 935		
Model	41.8377619	7	5.97682312	F(7, 927) = 44.75		
Residual	123.818521	927	.133569063	Prob > F = 0.0000		
Total	165.656283	934	.177362188	R-squared = 0.2526		
				Adj R-squared = 0.2469		
				Root MSE = .36547		

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0654307	.0062504	10.47	0.000	.0531642	.0776973
exper	.014043	.0031852	4.41	0.000	.007792	.020294
tenure	.0117473	.002453	4.79	0.000	.0069333	.0165613
married	.1994171	.0390502	5.11	0.000	.1227801	.276054
black	-.1883499	.0376666	-5.00	0.000	-.2622717	-.1144281
south	-.0909036	.0262485	-3.46	0.001	-.142417	-.0393903
urban	.1839121	.0269583	6.82	0.000	.1310056	.2368185
_cons	5.395497	.113225	47.65	0.000	5.17329	5.617704

The regression result includes estimated coefficients, standard errors, t -ratios p -values, and 95% confidence intervals.

The OLS point estimate $\hat{\beta}_1 = 0.0654307$. By its nature, it provides **no information** about how close $\hat{\beta}_1$ is “likely” to be β_1 , because β_1 is unknown.

- This limitation (i.e., β_1 is unknown.) can be “overcome” by constructing an interval to cover the unknown β_1 based on the sampling distribution of $\hat{\beta}_1$.
- $[\hat{\beta}_j - c * se(\hat{\beta}_j), \hat{\beta}_j + c * se(\hat{\beta}_j)]$ is called the 95% **confidence interval (CI)** for β_j .
- Suppose we have 100 random samples. For each sample, we can calculate $\hat{\beta}_j$ and $se(\hat{\beta}_j)$ to construct the confidence interval for the (unknown) population value β_j , $\hat{\beta}_j \pm c * se(\hat{\beta}_j)$. So, we will have 100 confidence intervals from these 100 samples. Then, β_j would lie in these confidence intervals for 95 times. In this sense, we say with probability .95, β_j is contained by 95% confidence $[\hat{\beta}_j \pm c * se(\hat{\beta}_j)]$.

- For the single sample, we do not know whether true β_j is actually contained in the confidence interval. We don't know whether this particular sample is among the 95%, even though this probability is .95. In the example of return to education, the 95% confidence interval for the partial effect of education on wage β_1 is [.0532, .0777]. If we believe the sample WAGE2 is among the 95% (the probability is .95), then β_1 lies in this interval [0.0532, 0.0777].

(C) Empirical Examples (Estimation, Interpretation, Testing)

- Consider the following multiple regression model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_K x_{Ki} + u_i, \quad i = 1, \dots, n$$

- Under the first 4 assumptions, $\hat{\beta}_0, \dots, \hat{\beta}_K$ are unbiased and consistent. With the additional assumption, i.e., the Assumption 5 ---- homoskedasticity, the OLS estimators are Best Linear Unbiased Estimators.
 - (1) Linear in parameters
 - (2) Random Samples
 - (3) No perfect collinearity
 - (4) Zero conditional mean
 - (5) Homoskedasticity
 - (6) Normality

Example: Return to Education (Wage2)

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{tenure} + \beta_4 \text{married} + \beta_5 \text{black} + \beta_6 \text{south} + \beta_7 \text{urban} + u$$

$$\hat{\beta} = (X'X)^{-1}X'y, \quad \widehat{\text{Var}}(\hat{\beta}_j) = \frac{\frac{1}{n-k-1} \sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (x_{ji} - \bar{x}_j)^2 (1-R_j^2)}, \quad \text{where } j = 1, \dots, k$$

. reg lwage educ exper tenure married black south urban

Source	SS	df	MS	Number of obs = 935		
Model	41.8377619	7	5.97682312	F(7, 927) = 44.75		
Residual	123.818521	927	.133569063	Prob > F = 0.0000		
Total	165.656283	934	.177362188	R-squared = 0.2526		
				Adj R-squared = 0.2469		
				Root MSE = .36547		

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0654307	.0062504	10.47	0.000	.0531642	.0776973
exper	.014043	.0031852	4.41	0.000	.007792	.020294
tenure	.0117473	.002453	4.79	0.000	.0069333	.0165613
married	.1994171	.0390502	5.11	0.000	.1227801	.276054
black	-.1883499	.0376666	-5.00	0.000	-.2622717	-.1144281
south	-.0909036	.0262485	-3.46	0.001	-.142417	-.0393903
urban	.1839121	.0269583	6.82	0.000	.1310056	.2368185
_cons	5.395497	.113225	47.65	0.000	5.17329	5.617704

Questions:

(a) How do you explain the R-squared value 0.2526 here?

(b) How do interpret $\hat{\beta}_{educ} = 0.065$?

(c) How do you interpret the 95% confidence interval for β_{educ} , i.e., [0.053,0.077]

(d) How do we test $H_0: \beta_{educ} = 0$ and $H_1: \beta_{educ} \neq 0$?

(e) Do you think that the estimator $\hat{\beta}_{educ}$ is an unbiased estimator? Why or why not?

(f) How do you fix or mitigate the problem found in (e)?

(g) Could you reduce the standard errors? If yes, how?

Endogeneity Problem

Example: Return to Education (WAGE2)

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married + \beta_5 black + \beta_6 south + \beta_7 urban + \beta_8 IQ + u$$

Source	SS	df	MS	Number of obs = 935		
Model	43.5360162	8	5.44200202	F(8, 926) = 41.27		
Residual	122.120267	926	.131879338	Prob > F = 0.0000		
Total	165.656283	934	.177362188	R-squared = 0.2628		
				Adj R-squared = 0.2564		
				Root MSE = .36315		

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0544106	.0069285	7.85	0.000	.0408133	.068008
exper	.0141458	.0031651	4.47	0.000	.0079342	.0203575
tenure	.0113951	.0024394	4.67	0.000	.0066077	.0161825
married	.1997644	.0388025	5.15	0.000	.1236134	.2759154
black	-.1431253	.0394925	-3.62	0.000	-.2206304	-.0656202
south	-.0801695	.0262529	-3.05	0.002	-.1316916	-.0286473
urban	.1819463	.0267929	6.79	0.000	.1293645	.2345281
IQ	.0035591	.0009918	3.59	0.000	.0016127	.0055056
_cons	5.176439	.1280006	40.44	0.000	4.925234	5.427644

We use *IQ* as the proxy variable for *ability*. What if we don't have suitable proxy variable for omitted variable *ability*?

Answer:

Besides using (a) the proxy variable approach, we can (b) use Instrumental Variables (IV) approach to “replace” *educ*, or (c) use Panel data model (e.g., fixed effect model) to directly eliminate *ability*.

Each approach requires **additional information**.

- (a) **Find** a proxy variable for *ability*
- (b) **Find** an IV for *educ*
- (c) **Re-interview** same group of people more than one period

Questions: How do we know the homoskedasticity assumption is satisfied? Why is it so important to answer this question?

Under the null homoskedasticity, LM statistic = $n \times R_{\hat{u}^2}^2 \sim \chi_k^2$ (Breusch-Pagan test)

If $LM > \chi_k^2$, we reject the null. Therefore, heteroskedasticity is present.

$$\hat{u}^2 = \gamma_0 + \gamma_1 educ + \gamma_2 exper + \gamma_3 tenure + \gamma_4 married + \gamma_5 black + \gamma_6 south + \gamma_7 urban + \gamma_8 IQ + v$$

```
. reg uhat2 educ exper tenure married black south urban IQ
```

Source	SS	df	MS			
Model	1.44972973	8	.181216216	Number of obs =	935	
Residual	54.7117297	926	.059083941	F(8, 926) =	3.07	
Total	56.1614595	934	.060130042	Prob > F =	0.0020	
				R-squared =	0.0258	
				Adj R-squared =	0.0174	
				Root MSE =	.24307	

uhat2	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
educ	-.0018405	.0046375	-0.40	0.692	-.0109417	.0072608
exper	-.0009025	.0021185	-0.43	0.670	-.0050602	.0032552
tenure	-.006821	.0016328	-4.18	0.000	-.0100254	-.0036167
married	.0065477	.025972	0.25	0.801	-.0444232	.0575186
black	-.0219303	.0264339	-0.83	0.407	-.0738075	.0299469
south	.0312713	.0175721	1.78	0.075	-.0032145	.0657571
urban	.0081137	.0179335	0.45	0.651	-.0270814	.0433087
IQ	.0005953	.0006639	0.90	0.370	-.0007076	.0018981
_cons	.1353629	.0856758	1.58	0.114	-.0327784	.3035042

LM statistic = $n \times R_{\hat{u}^2}^2 = 935 \times 0.0258 \approx 24.12 > \chi^2(8) = 2.773$. We reject the null at 5% significance level because the critical value is 2.773.

We reject the assumption 5: Homoskedasticity. Using robust standard estimation procedure, we can obtain the estimated robust standard errors: the corrected standard errors.

```
. reg lwage educ exper tenure married black south urban IQ, r
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Linear regression

```
Number of obs =    935
F(  8,    926) =   48.51
Prob > F       =   0.0000
R-squared      =   0.2628
Root MSE      =   .36315
```

lwage	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
educ	.0544106	.007273	7.48	0.000	.0401372	.068684
exper	.0141458	.0032382	4.37	0.000	.0077908	.0205009
tenure	.0113951	.0025368	4.49	0.000	.0064166	.0163736
married	.1997644	.0390896	5.11	0.000	.1230499	.2764788
black	-.1431253	.0376459	-3.80	0.000	-.2170064	-.0692442
south	-.0801695	.0277381	-2.89	0.004	-.1346063	-.0257327
urban	.1819463	.0267419	6.80	0.000	.1294646	.234428
IQ	.0035591	.0009564	3.72	0.000	.0016822	.0054361
_cons	5.176439	.1212236	42.70	0.000	4.938534	5.414344

- Remarks:

(a) Even if we reject the homoskedasticity assumption, we usually do not know the structure of heteroskedasticity.

(b) We could use the robust standard errors (especially when we have a fairly large sample size). Even if there is no heteroskedasticity, so long as the sample size is **large** enough we can still use the robust standard errors as they are appropriate even under homoskedasticity.

(c) Usually, the robust standard errors are bigger than the uncorrected wrong standard errors.

Appendix

To derive 95% confidence interval formula in Section (B), we start with the fact that

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim N(0,1)$$

- We obtain,

$$P\left(-1.96 < \frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} < 1.96\right) = .95$$

- Equivalently,

$$P(\hat{\beta}_j - 1.96 * sd(\hat{\beta}_j) < \beta_j < \hat{\beta}_j + 1.96 * sd(\hat{\beta}_j)) = .95$$

- It says that the interval

$$[\hat{\beta}_j - 1.96 * sd(\hat{\beta}_j), \quad \hat{\beta}_j + 1.96 * sd(\hat{\beta}_j)]$$

covers the unknown population β_j with probability 0.95. This is also called the **interval estimator** of β_j .

- Since $sd(\hat{\beta}_j)$, the standard deviation of $\hat{\beta}_j$, involves the unknown σ^2 , we can replace $sd(\hat{\beta}_j)$ with standard error of $\hat{\beta}_j$, $se(\hat{\beta}_j)$. Then,

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$$

- It follows that

$$P\left(c_{2.5} < \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} < c_{97.5}\right) = .95$$

or

$$P(\hat{\beta}_j - c_{97.5} * se(\hat{\beta}_j) < \beta_j < \hat{\beta}_j - c_{2.5} * se(\hat{\beta}_j)) = .95$$

where $c_{2.5}$ and $c_{97.5}$ are the 2.5th percentile and 97.5th percentile of t distribution with $n - k - 1$ degrees of freedom, respectively. In addition, $|c_{2.5}| = c_{97.5} = c$. When the degree of freed is large, $c_{2.5}$ and $c_{97.5}$ equal to -1.96 and 1.96 respectively.

- $[\hat{\beta}_j - c * se(\hat{\beta}_j), \hat{\beta}_j + c * se(\hat{\beta}_j)]$ is called the 95% **confidence interval (CI)** for β_j .